Lab: Predicting churn with an LLM

No code required · ~50 minutes · 5 prediction runs + 1 comparison
We have built a ML churn model in code. In this exercise we'll make predictions using a LLM, and find out how much we could trust them.

You need **two different AI chatbots** (any two of ChatGPT, Claude, Gemini — free versions are fine) and these three files here https://github.com/pearl-yu/mist5400fall2026/tree/main/week3/llm_pred_data :

`lab_customers.csv`	The 45 customers to predict — no answers included
`examples_20.csv`	20 past customers whose outcome is known
`lab_ground_truth.csv`	Added actual target values to lab_customers.csv

The rules — they make the experiment valid
1. Fresh chat for every run. Never reuse a conversation. Earlier messages contaminate later answers.
2. Record the exact model name. It's shown in the model picker at the top of the chat (e.g. "GPT-5 Thinking", "Claude Sonnet 4.5", "Gemini 2.5 Flash"). "ChatGPT" is not a model name.
No file upload on your chatbot's free tier? Open the CSV in any text editor, copy everything, and paste it at the end of the prompt instead.

Experiment 1 — Zero-shot, twice (2 conversations)

Fresh chat on your main chatbot.
Drag in: `lab_customers.csv` to the conversation *only*.

Then paste in this prompt:

```
The attached file lab_customers.csv contains 45 telecom customers. For EACH
customer, predict whether they will churn (Yes or No) and give your probability
that they churn (a number between 0 and 1). tenure = months as a customer;
MonthlyCharges = monthly bill in USD.

Do NOT write or run code — read the rows and predict directly yourself.

Write the answers to a CSV file with exactly 45 lines and no header, one line per
customer, in this format:
customer_id,prediction,probability
```

```
No explanation, no other text.
```

Save the output as **pred_1a_<modelname>_<lastname>.csv**

Now start a new conversation, upload the same csv file, same prompt. Save the output as Save as **pred_1b_<modelname>_<lastname>.csv**

Experiment 2 — Few-shot examples, twice (2 conversations)

Fresh chat, same chatbot.
Drag in: **lab_customers.csv** and **examples_20.csv**.

Then paste this prompt:

```
The attached file lab_customers.csv contains 45 telecom customers. For EACH
customer, predict whether they will churn (Yes or No) and give your probability
that they churn (a number between 0 and 1). tenure = months as a customer;
MonthlyCharges = monthly bill in USD.

The second attached file contains real customers from the same company whose
outcome is already known (see the Churn column). Use these labeled examples to
calibrate your predictions.

Do NOT write or run code — read the rows and predict directly yourself.

Write the answers to a CSV file with exactly 45 lines and no header, one line per
customer, in this format:
customer_id,prediction,probability
No explanation, no other text.
```

Save as **pred_2a_<modelname>_<lastname>.csv**.

Then run the identical setup once more in a **new fresh chat** and save as **pred_2b_<modelname>_<lastname>.csv**

Experiment 3 — A different model (1 conversation)

Run the Experiment 2 setup (both files, same prompt) once on your **second chatbot** (different company, or a different model in the picker). Fresh chat. Save as **pred_3_<modelname>_<lastname>.csv**

Now Upload all five prediction files to predictions.

Final task — Evaluation & Agreement across models (1 conversation)

Fresh chat on your main chatbot. **Drag in: all five of your saved CSV files, and `lab_ground_truth.csv`**

Then paste this prompt:

```
Attached are 5 prediction files for the same 45 customers, each produced in a
separate conversation. pred_1a and pred_1b: same model, no examples. pred_2a and
pred_2b: same model, with labeled examples. pred_3: a different model, with
labeled examples. The file names contain the LLM model used. Also
lab_ground_truth.csv that contains the true target variable values. (churn or
not).

Visualize the prediction accuracies and ROC.

Compare the five files. I need three comparisons: (1)Same-model reproducibility:
Do predictions disagree and by how much? (2)Few-shot effect: Did adding labeled
examples change the predictions (experiment1 vs experiment2)? (3) Cross-model
agreement: Did using a different model change predictions (experiment 2 v.s.
experiment)
```

Read the answer. Was your chatbot as consistent as you guessed?

> Copy the full response into a word document (or ask your AI tool to give you a PDF report), upload to **`LLM_pred In-class exercises`** in eLc

Now as Pearl has your prediction files, she'll check later for the same customer, how much the predictions vary from the same prompt. We'll discuss the results and if we use LLMs for predictions, what we need to look at to evaluate the robustness of results.